Series circuits

antocciphysics | TEACHER KEY | CPA

Learning target: Use current continuity and voltage drops in a single-path circuit.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 4; 3 course days later.*
5 J of electric potential-energy difference per coulomb.

02. Explain a worked example (worked)
Same 2 A through both; larger resistor has larger voltage drop.

03. Look and explain (model)
All current stops in the ideal single loop.

04. Build the solution (practice)
R_{total}=6 Ω; I=1 A.

05. Use the idea (reason)
Steady series current must be equal; unequal currents would change charge stored at the connection.

06. Choose a method (mixed) *Review: Unit 6, Day 13; 12 course days later.*
25° from the normal, on the other side.

07. Exit check (exit)
6 V.

08. STRETCH: Apply it another way (stretch)
Example 4+6 ohms: 1 A, drops 4+6 V; 2+8 ohms also works.

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01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 4; 3 course days later.*
21 J and 35 J; total 56 J.

02. Explain a worked example (worked)
Same 2 A through both; larger resistor has larger voltage drop.

03. Represent and interpret (model)
All current stops in the ideal single loop.

04. Solve and justify (practice)
30 Ω; .40 A; 4 V and 8 V.

05. Analyze the reasoning (reason)
Steady series current must be equal; unequal currents would change charge stored at the connection.

06. Choose a method (mixed) *Review: Unit 6, Day 13; 12 course days later.*
Image moves 0.4 m toward the mirror from behind it. Separation decreases by 0.8 m.

07. Exit check (exit)
New resistance 3 R versus 2 R; current becomes 2/3 of the original.

08. STRETCH: Apply it another way (stretch)
Total 10 ohms, R 5. Powers I^2R: 8, 12, 20 W; sum 40 W.